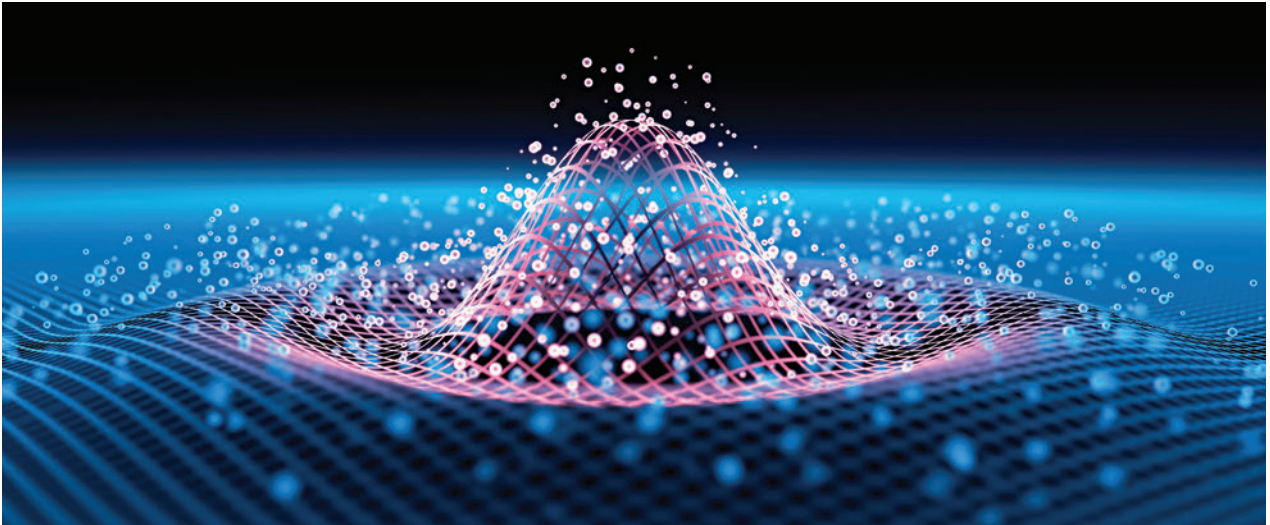




quantum bits that hold information in a unique way. Qubits can be both 0 and 1 at the same time, like a spinning coin in the air. This “superposition” ability unlocks powerful calculations, but it’s also incredibly fragile.

Decoherence is like the breeze in our card tower. It describes how a qubit loses its delicate superposition when it interacts with its surroundings. Even tiny vibrations, temperature changes, or even light can “decohere” the qubit, forcing it to choose just one state (0 or 1) and losing its special power.

Think of a music track on a record player. The tiny groove tells the needle exactly what sound to play. But if the record gets dusty or scratched, the needle can’t read it properly, introducing noise and distorting the music.

Noise in quantum computers comes from similar sources. It’s any unwanted influence that messes up the calculations. Just like the scratches on the record, noise can flip qubits from 0 to 1 or vice versa, introducing errors into the results.

These problems are big hurdles for building reliable quantum computers. If the calculations are full of errors, they become useless. However, scientists are working on ways to overcome these challenges.

Quantum Computers from Leading Companies

The race to unlock the immense potential of quantum computing is on, with several leading companies and startups taking unique approaches. Let’s explore the efforts of some key players in this exciting field:

IBM Quantum Experience

- **Publicly Accessible Qubits:** IBM boasts the most extensive cloud-based platform for quantum computing, the IBM Quantum Experience. Offering access to real quantum processors with up to 127 qubits (as of February 2024), it allows

anyone to experiment with writing and running quantum programs. This democratisation of access helps drive research and fosters a strong quantum community.

- **Qiskit Programming Framework:** To simplify quantum programming, IBM developed Qiskit, an open-source framework with tools for designing, simulating, and running quantum circuits. Qiskit’s user-friendly interface and compatibility with major cloud platforms make it a popular choice for researchers and developers alike.

Google Quantum AI

- **Quantum Supremacy:** Google grabbed headlines in 2019 by achieving “quantum supremacy” with their Sycamore processor. This milestone demonstrated a quantum computer performing a calculation impossible for any classical computer in a practical timeframe. They continue pushing the boundaries with their latest 53-qubit processor, “Osprey.”
- **Superconducting Qubits:** Google primarily focuses on superconducting qubits, which leverage extremely cold temperatures to maintain their delicate quantum states. This technology offers scalability and has shown promising results in various applications, including materials science and drug discovery.

Intel Quantum Computing

- **Beyond Superconducting:** Intel takes a multi-pronged approach, exploring both superconducting and spin qubits. Spin qubits, utilising the quantum properties of electron spins, are potentially more fault-tolerant but face challenges in scalability. This exploration of diverse technologies allows Intel to hedge its bets and potentially discover breakthrough solutions.
- **Focus on Error Correction:** Recognizing the critical challenge of decoherence, Intel invests heavily in developing